We look forward to seeing the results of applying these various methods in an actual clinical series. More importantly, as we indicate in our article, the ultimate goal is the development of accurate methods of measuring true corneal power.

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Dear Editor:

I offer some insights regarding the data presented by Hamed et al[1] that should allow the application of this information for all cataract surgeons in a practical manner.

Although the clinical history method of determining the true central corneal refractive power was found to be the most accurate way of determining the true refractive power of the cornea,[1–3] this method requires that the keratometry values before the LASIK procedure be available. Since the time between the LASIK and cataract procedures will often extend to more than 10 years and the LASIK and the cataract surgery will often be performed by different surgeons, pre-LASIK keratometry will be difficult and sometimes impossible to obtain for many patients.

The contact lens method has also been proposed as an accurate way of determining the true corneal refractive power.[1–3] This technique is cumbersome to perform, which makes practical application of this technique in a busy cataract practice difficult. It is my experience that this technique is seldom used.

Computerized videokeratography can also be used to estimate the corneal refractive power; however, Hamed et al[1] have demonstrated that this will produce misleading values, as it uses a standardized corneal refractive index that is known to have been changed by the refractive surgery. Although adjustments in this value may increase the accuracy of this value,[4] the variety of corneal topography machines available will lead to confusion regarding the correct adjustment factors for each machine.

Based on the results of Hamed et al and my personal experience with this problem over the last 8 years, I offer the following recommendations for the adjustment of the keratometry value and hence the intraocular lens (IOL) power after LASIK:

1. The IOL power is calculated in the standard manner utilizing the keratometry values and the axial length.
2. The amount of LASIK refractive correction is obtained from the patient or medical records. This value is usually easier to obtain, as patients may be aware of their previous correction or have an old pair of glasses, or old LASIK records may be available.
3. Subtract 1 diopter from the mean measured readings for every 4 D of myopic LASIK correction. This is based on the assumption that the true central keratometry value will be approximately 75% less than the measured keratometry value. This percentage reflects the results of previous studies indicating that the 3.0-mm keratometry values underestimate the true refractive change at the cornea by 24%[5] and therefore indicate only 76% of the induced corneal flattening.
4. Aiming for a postoperative refractive goal of −0.50 to −1.0 D would introduce another margin of safety to avoid postoperative hyperopia but would also increase the LASIK enhancement rate. Postoperative myopia is a much preferred postoperative outcome as it will be better tolerated by the patients and, if necessary, much more easily corrected by a LASIK enhancement.
5. Patients should be counseled preoperatively that a small LASIK enhancement may be required after the cataract procedure.

Although these recommendations involve estimations, they should provide an easily applied practical method that all cataract surgeons can use with their existing equipment and formulas to achieve excellent postoperative cataract refractive results in most cases while avoiding the highly undesirable postoperative hyperopia surprise.

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Author reply

Dear Editor:

We thank Drs. Probst and Holladay for their comments. However, we are somewhat confused by their recommendations. In point 1, they recommend calculating intraocular lens power in the standard manner using the measured keratometry values. Then, in point 3, they recommend using readings that have been modified by subtracting 1.0 D for every 4.0 D of myopic LASIK correction. In fact, this latter recommendation is nearly identical to what we have recommended in our article, in which we suggest that readings be reduced by 0.24 D for every diopter of myopic LASIK correction. Therefore, for 4.0 D of LASIK correction, the keratometry reading should be reduced by 0.96 D.

Unfortunately, as we point out in our article, using keratometry readings is not as accurate as using the EffRP value that we determine from the EyeSys device (EyeSys Vision, Houston, Texas). Therefore, we hold to our recommendation to use the EffRP values whenever they are available. The keratometry values can be used if the EyeSys device is not available for measuring the patient.